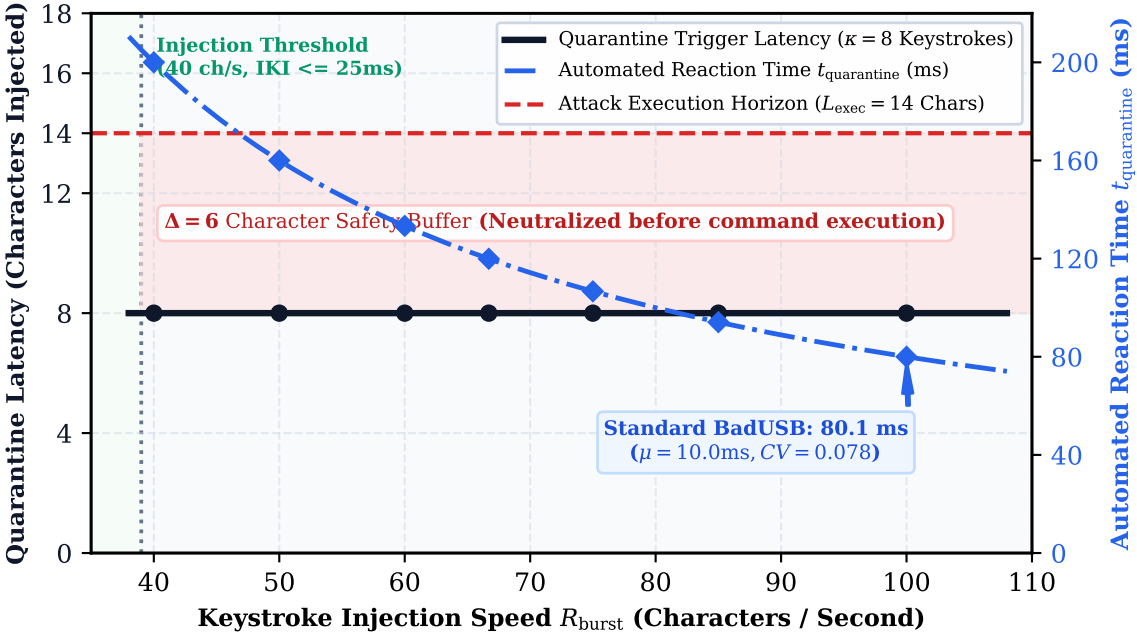
Quarantine Trigger Latency ($\kappa = 8$ Keystrokes)
Automated Reaction Time $t_{\text{quarantine}}$ (ms)
Attack Execution Horizon ($L_{\text{exec}} = 14$ Chars)
Injection Threshold
(40 ch/s, IKI <= 25ms)
$\Delta = 6$ Character Safety Buffer (Neutralized before command execution)
Standard BadUSB: 80.1 ms
($\mu = 10.0$ms, $CV = 0.078$)
Quarantine Latency (Characters Injected)
Automated Reaction Time $t_{\text{quarantine}}$ (ms)
Keystroke Injection Speed R_{burst} (Characters / Second)
0
4
8
12
14
16
18
0
40
80
120
160
200
40
50
60
70
80
90
100
110